

Dear Students,

below are the group projects, each with specific requirements that must be met. Parameters not explicitly specified should be selected based on the given application (e.g., switching frequency, capacitor sizing, and so forth). In all cases, the junction temperature must not exceed 100 °C and the power factor should be assumed in the range (0.8;1). Efficiency calculation should be performed considering solely power switch losses and auxiliary losses (i.e., ignore capacitor and magnetics losses).

The following key waveforms must be included:

- Pole, line-to-line, phase voltages;
- Phase current;
- Spectral analysis of both phase voltage and phase current;
- DC-link capacitor voltages.

Additionally, the following quantity must be included:

- Total Harmonic Distortion (THD) of the voltage;
- Efficiency curve must be plotted from 10 % to 100 % of the rated power;
- 4-point efficiency and European efficiency;
- Losses breakdown among the various switches.

The report should be performed with reference to the Overleaf template (check in Virtuale). It must be delivered at least two days before the exam date (in case of midterms, two days before the second midterm). In case report authors will attend different exam dates, the report submission deadline is the one related to the earliest exam. The report in PDF format is submitted using the form link (will be available soon in Virtuale). PLECS files and any supporting materials should be delivered in zip format.

Group 1

Application: 20 kW BESS (Battery Energy Storage System) inverter for the USA market assuming ambient temperature 25 °C

Topology: three-phase three-level NPC inverter

Modulation technique: sinusoidal PWM

DC voltage: 1500 V

Line-to-line AC voltage: 480 V

Power switch technology: SiC-MOSFET

Group 2

Application: 40 kW BESS (Battery Energy Storage System) inverter for the European market assuming ambient temperature 25 °C

Topology: three-phase three-level NPC inverter

Modulation technique: sinusoidal PWM

DC voltage: 1000 V

Line-to-line AC voltage: 400 V

Power switch technology: Si-IGBT

Group 3

Application: 5 kW three phase PV inverter in the western region of Japan assuming ambient temperature 30 °C

Topology: three-phase three-level NPC inverter

Modulation technique: sinusoidal PWM

DC voltage: 600 V

Line-to-line AC voltage: 200 V

Power switch technology: Si-IGBT

Group 4

Application: 10 kW automotive inverter systems for EV charger applications in European market assuming ambient temperature 30 °C

Topology: three-phase three-level T-type inverter common emitter configuration

Modulation technique: sinusoidal PWM

DC voltage: 800 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 5

Application: 7.5 kW solar-fed residential inverter in Indian market assuming ambient temperature 30 °C

Topology: three-phase three-level flying capacitor inverter

Modulation technique: Level-shift sinusoidal PWM

DC voltage: 700 V

Line-to-line AC voltage: 415 V

Power switch technology: SiC-MOSFET

Group 6

Application: 12.5 kW commercial building inverter for smart-grid systems in Europe assuming ambient temperature 40 °C

Topology: three-phase three-level flying capacitor inverter

Modulation technique: Phase-shift sinusoidal PWM

DC voltage: 800 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 7

Application: 10 kW photovoltaic inverter for the North-European market assuming ambient temperature 25 °C

Topology: three-phase two-level inverter

Modulation technique: SVPWM (Space Vector PWM - centered)

DC voltage: 800 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 8

Application: 15 kW grid-tied inverter system in the USA market assuming ambient temperature 25 °C

Topology: three-phase three-level T-type inverter common collector configuration

Modulation technique: sinusoidal PWM

DC voltage: 900 V

Line-to-line AC voltage: 480 V

Power switch technology: SiC-MOSFET

Group 9

Application: 20 kW BESS (Battery Energy Storage System) inverter for the USA market assuming ambient temperature 25 °C

Topology: three-phase three-level CHB inverter

Modulation technique: Level-shift sinusoidal PWM

DC voltage: 1500 V

Line-to-line AC voltage: 480 V

Power switch technology: SiC-MOSFET

Group 10

Application: 40 kW photovoltaic inverter for the South-European market assuming ambient temperature 30 °C

Topology: three-phase two-level inverter

Modulation technique: SVPWM (Space Vector PWM - centered)

DC voltage: 1500 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 11

Application: 15 kW three phase inverter with 16.7 Hz AC power supply in the Germany railway traction system assuming ambient temperature 40 °C

Topology: three-phase three-level flying capacitor inverter

Modulation technique: phase-shift sinusoidal PWM

DC voltage: 1000 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 12

Application: 6 kW BESS (Battery Energy Storage System) inverter for the Singapore market assuming ambient temperature 30 °C

Topology: three-phase three-level flying capacitor inverter

Modulation technique: Level-shift sinusoidal PWM

DC voltage: 650 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 13

Application: 22 kW on-board EV charger in the China market assuming ambient temperature 35 °C

Topology: three-phase three-level flying capacitor inverter

Modulation technique: Level-shift sinusoidal PWM

DC voltage: 1000 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET

Group 14

Application: 20 kW industrial motor drive inverter at 300 Hz assuming ambient temperature 35 °C

Topology: three-phase three-level T-type inverter common emitter configuration

Modulation technique: sinusoidal PWM

DC voltage: 1000 V

Line-to-line AC voltage: 600 V

Power switch technology: SiC-MOSFET

Group 15

Application: 3.3 kW photovoltaic inverter for the USA market assuming ambient temperature 25 °C

Topology: three-phase two-level inverter

Modulation technique: sinusoidal PWM

DC voltage: 400 V

Line-to-line AC voltage: 208 V

Power switch technology: Si-IGBT

Group 16

Application: 10 kW residential PV for the Canadian market assuming ambient temperature 25 °C

Topology: three-phase two-level inverter

Modulation technique: DPWM1

DC voltage: 380 V

Line-to-line AC voltage: 208 V

Power switch technology: Si-IGBT

Group 17

Application: 3.5 kW onboard aircraft inverter with 400 Hz AC power supply assuming ambient temperature 5 °C

Topology: three-phase two-level inverter

Modulation technique: DPWM2

DC voltage: 350 V

Line-to-line AC voltage: 199 V

Power switch technology: SiC-MOSFET

Group 18

Application: 40 kW BESS (Battery Energy Storage System) inverter for the European market assuming ambient temperature 25 °C

Topology: three-phase three-level T-type inverter common collector configuration

Modulation technique: sinusoidal PWM

DC voltage: 1000 V

Line-to-line AC voltage: 400 V

Power switch technology: Si-IGBT

Group 19

Application: 10 kW three phase inverter with 16.7 Hz AC power supply in the German railway traction system assuming ambient temperature 10 °C

Topology: three-phase two-level inverter

Modulation technique: SVPWM (Space Vector PWM - centered)

DC voltage: 800 V

Line-to-line AC voltage: 400 V

Power switch technology: SiC-MOSFET